Large Prime Gaps: Adjacent-Dimension Cancellation, Finite Moment Certificates, and Density Transfer

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Abstract

We study the signed smooth-divisor-square construction in *Positive lower density of large prime gaps*. Its prime-deletion identity couples consecutive function dimensions, including dimension zero. We give a finite, admissible construction with truncated Poisson level weights that cancels every interior uncut term exactly. For a fixed compactly supported profile, the remaining normalized marked energy is exponentially small in the maximum dimension. A rational Dirichlet taper gives a simpler polynomial-rate alternative. We prove support-preserving approximation and moment-matrix error bounds and an explicit weighted-to-ordinary density ledger. The Poisson distribution here describes chosen coefficients, not a probabilistic assertion about primes. The source's uniform mixed-moment and singular-series arguments are mapped and read, but their global arithmetic conclusions remain imported rather than independently certified by these local results. Private original moment and approximation arguments are applied with their finite-dimensional and common-measure restrictions intact. Contribution assessment: C3/5, exponential finite-family cancellation and explicit stability and density budgets notably sharpen local source inputs without certifying the global arithmetic theorem.

Contents

1 Scope, quantifiers, and the source architecture **2**

2 The exact analytic problem **3**

3 Profiles and quantitative cutoff control **4**

4 Exponential finite-family cancellation **5**

5 A rational taper as an alternative **6**

6 Moment coordinates with an error certificate **7**
6.1 The operator and its conditioning 7
6.2 Exact finite representations and cross-dimension moments 8
6.3 An admissible approximation mechanism 9

7 A finite weighted-to-prime density ledger **9**

8 The arithmetic boundary is still substantive **10**

9 Comparison, costs, and further work **11**
9.1 Current public context 11
9.2 Advantages and costs 12
9.3 Feasible next questions 12

10 Evidence and contribution assessment **12**

1 Scope, quantifiers, and the source architecture

Let p_n denote the nth prime and $d_n = p_{n+1} - p_n$. The source [1] asserts that, for every fixed $C > 0$, there exist $c(C) > 0$ and $N_0(C)$ such that

$$\#\{n \le N : d_n > C \log p_n\} \ge c(C)N \qquad (N \ge N_0(C)). \tag{1}$$

This is ordinary lower density in the prime index, for every sufficiently large N. A subsequence of large gaps, a weighted density, or a positive proportion of prime-free intervals starting at arbitrary integers is not the same statement. Many empty integer-start intervals can lie in one exceptionally long gap. No random-sieve model is substituted for the primes in this report.

The complete assigned build consists of six TeX files and one bibliography: 1,255 TeX lines and 215 bibliography lines. All 1,470 lines were displayed in bounded, untruncated reads. This is a reading claim, not a claim to have independently reproved every analytic-number-theory input. The following map follows the manuscript's input order; its numerically named moments file is placed last in the source.

File and lines	Mathematical role and audit boundary
`main.tex`, 44	Assembly and notation; all section inputs mapped.
`01-introduction.tex`, 137	Lower-density claim, comparison with integer-start gaps, and fixed-parameter order.
`02-counting.tex`, 144	Weighted detection, removal of weights, and bounded multiplicity; finite forms proved below.
`04-weights.tex`, 131	Cumulative functions, exact deletion, and moment contracts; deletion and local forms checked below.
`05-cancellation.tex`, 181	Small-mean profile, flat level window, and cutoff control; locally replaced by new finite choices.
`03-moments.tex`, 618	CRT, distribution in progressions, Fourier/Euler products, singular-series averaging, and moment assembly; architecture and hypotheses checked, global arithmetic imported.
`references.bib`, 215	Bibliographic data, not a reading of all cited papers.

The original private proof reading covers 295 distinct lines in three files, not the full private corpus: [5], lines 121–181 and 455–477; [6], lines 216–260 and 305–328; and [7], lines 14–91 and 163–226. Their respective roles are metric-aware finite moment reconstruction, same-law second moments with remainder control, and Taylor moment expansions with normalization cautions. Included chapter files were not followed. Literal intervals, unread complements, and matching pre/post hashes are recorded in the owned read-scope ledger. Snapshot authentication supplied separately by the project is provenance, not mathematical verification or a live-branch claim.

2 The exact analytic problem

Fix $\lambda > 0$, let X tend to infinity through integers, and put

$$L = \log X, \quad h = \lfloor \lambda L \rfloor, \quad J = \{1, \dots, h\}, \quad I = \{h+1, \dots, 2h\}, \quad \tau = \tfrac{1}{8}.$$

The average $\mathbb{E}_X$ is $X^{-1} \sum_{X < m \le 2X}$, and

$$\vartheta(n) = (\log n) \mathbf{1}_{\{n \text{ prime}\}}, \qquad V_A(m) = L^{-1} \sum_{a \in A} \vartheta(m+a).$$

For $A = I, J$, either $V_A = 0$ or $V_A \ge 1$, since $m + a > X$. All function norms below are real Lebesgue L^2 norms on the nonnegative orthant, with absolute value as the dimension-zero norm.

For a positive integer k choose symmetric real

$$f_j \in C_c^\infty(\Delta_j(\tau)), \qquad \Delta_j(\tau) = \{t_i > 0 : \textstyle\sum_i t_i < \tau\}, \quad 1 \le j \le k,$$

allow $f_0 \in \mathbb{R}$, and put $f_{k+1} = 0$. Define

$$F_j(v) = \int_{s_i \ge v_i} f_j(s)\, ds, \quad F_0 = f_0, \qquad (T f_{j+1})(t) = \int_0^\infty f_{j+1}(t,u)\, du.$$

Then $(-1)^j \partial_1 \cdots \partial_j F_j = f_j$. It is this complete derivative that vanishes on coordinate faces; F_j itself usually does not. This distinction is essential to the unmatched-shift argument. Write μ_{Mob} for the Möbius function and

$$D_F(m; b_1, \dots, b_j) = \sum_{d_i \mid m + b_i} \prod_i \mu_{\text{Mob}}(d_i) F\left(\frac{\log d_1}{L}, \dots, \frac{\log d_j}{L} \right).$$

Dimension zero is a scalar. Symmetry permits unordered shift sets. Set

$$\begin{aligned} Z(m) &= \sum_{j=0}^{k} \sum_{\substack{S \subset I \\ |S| = j}} D_{F_j}(m; S), \qquad \alpha_j = \frac{\lambda^j}{j!}, \\ w(f) &= \sum_{j=0}^{k} \alpha_j \left\| f_j \right\|_2^2, \qquad v(f) = \lambda \sum_{j=0}^{k} \alpha_j \left\| f_j + T f_{j+1} \right\|_2^2. \end{aligned} \tag{2}$$

Proposition 2.1 (Prime deletion, source identity)**.** *If $a \in I$ and $m + a$ is prime, then*

$$Z(m) = \sum_{j=0}^{k} \sum_{\substack{U \subset I \setminus \{a\} \\ |U| = j}} D_{F_j + F_{j+1}(\,\cdot\,, 0)}(m; U).$$

The signed complete derivative in dimension j of the new function is $f_j + T f_{j+1}$, with no factor $j+1$.

Proof. For a subset containing a, the divisor in that coordinate is either 1 or $m + a$. The latter has logarithmic coordinate greater than $1 > \tau$ and hence gives zero. Deleting that coordinate is a bijection from subsets containing a to subsets of $I \setminus \{a\}$ of one lower cardinality. Subsets not containing a supply F_j. Differentiating the cumulative integrals gives $f_j + \int f_{j+1}(\,\cdot\,, u)\, du$. For $j = 0$ this says exactly $f_0 + \int f_1$. □

Imported arithmetic contract. Source Proposition `prop:weight-moments`, with λ, k, f fixed before $X \to \infty$, states

$$\begin{aligned} \mathbb{E}_X Z^2 &\to w, \qquad \mathbb{E}_X Z^4 = O_{\lambda,k,f}(1), \\ \mathbb{E}_X Z^2 V_J &\to \lambda w, \qquad \mathbb{E}_X Z^2 V_I \to v, \\ \limsup_{X\to\infty} \mathbb{E}_X Z^2 V_J^2 &\le C_* w, \qquad C_* = \lambda + \lambda^2 c_G^2, \quad c_G = \int_0^\infty |G'(u)|^2 \, du. \end{aligned} \tag{3}$$

Here $G \in C_c^\infty(\mathbb{R})$ is fixed, $G(0) = 1$, and $G(u) = 0$ for $u \ge \tau$. The important uniformity is that C_* is independent of k, f. The fourth-moment bound is *not* stated uniformly in them. When $w > 0$, $W_X = Z^2/w$ is nonnegative, has limiting mean one, and has right-block marked mean v/w. We improve this last analytic ratio, not the arithmetic contract itself.

3 Profiles and quantitative cutoff control

Lemma 3.1 (Source profile construction)**.** *For every $\lambda > 0$ and every $\eta > 0$ there is a nonnegative $g \in C_c^\infty((0,\infty))$ such that*

$$\int g^2 = 1, \qquad \int g = \sqrt{\lambda}, \qquad \mu_g := \int u g(u)^2 \, du < \eta.$$

Its support is contained in $[b, B]$ for some $0 < b < B < \infty$. Neither B nor derivative bounds are asserted uniform as η or λ varies.

Proof. Choose a smooth ϕ_R, supported in $(1/2, 2R)$, equal to $1/u$ on $[1, R]$, and between zero and $1/u$ everywhere. Let $A_R = \int \phi_R$, $N_R^2 = \int \phi_R^2$, and $B_R = \int u\phi_R^2$. Then $A_R \ge \log R$, $B_R \le \log(4R)$, and N_R is positive and bounded for $R \ge 2$. With $g_0 = \phi_R/N_R$ and $c = \lambda/(\int g_0)^2$, take $g(u) = c^{-1/2} g_0(u/c)$. Direct changes of variable give the first two identities and $\mu_g = \lambda B_R/A_R^2 \to 0$. All support and smoothness claims hold for each fixed finite R. □

Lemma 3.2 (Bounded-profile concentration)**.** *Let U_i be independent with common density $g(u)^2$, supported in $[0, B]$ and of mean μ_g. For $\rho > 0$ put $S_j = (U_1 + \cdots + U_j)/\rho$, with $S_0 = 0$. If $j \ge 1$ and $a > j\mu_g/\rho$, then*

$$\mathbb{P}(S_j > a) \le \exp\left(-\frac{2\rho^2 (a - j\mu_g/\rho)^2}{jB^2} \right). \tag{3.1}$$

Proof. For any probability law on $[0, B]$ its variance is at most $B^2/4$: variance is the minimum squared error about a constant and is at most $\mathbb{E}(U - B/2)^2$. The second derivative of $H(t) = \log \mathbb{E} e^{t(U - \mathbb{E}U)}$ is the variance under its exponentially tilted law, also supported in $[0, B]$. Thus $H(0) = H'(0) = 0$ and $H(t) \le t^2 B^2/8$ for $t \ge 0$. Independence, Markov's inequality, and minimization at $t = 4(\rho a - j\mu_g)/(jB^2)$ prove the claim. □

Fix $0 < \beta < \gamma < \tau$, and choose a smooth real cutoff $0 \le \chi \le 1$, equal to one on $[0, \beta]$ and zero on $[\gamma, \infty)$. For any $\rho > 0$ define

$$Q_j^\rho(t) = \rho^{j/2} \prod_{i=1}^{j} g(\rho t_i), \quad Q_0^\rho = 1, \qquad T Q_{j+1}^\rho = \sqrt{\lambda/\rho}\, Q_j^\rho. \tag{4}$$

These have norm one. Integration against $(Q_j^\rho)^2 dt$ is expectation with respect to the law of S_j just defined. In contrast, the deletion integral uses the probability law $g(u)\, du/\sqrt{\lambda}$. These two laws are generally different. We control the latter by the deterministic bound $u \le B$, not by silently replacing it with the squared-profile law.

4 Exponential finite-family cancellation

The following construction uses all levels, including the source's permitted scalar level zero. It is a local improvement of the flat window in `05-cancellation.tex`. The factorial recurrence and exponential-moment bounds are classical; no historical novelty of those principles is claimed.

Theorem 4.1 (Truncated Poisson level construction)**.** *Fix $\lambda > 0$, $0 < \theta < 1$, and $0 < \beta < \gamma < \tau$. Choose the profile in Lemma 3.1 so that $\mu_g < \theta\beta$. Put $d = \beta - \mu_g/\theta > 0$, and bound* $\operatorname{supp} g \subset (0, B]$. *For every integer $k \geq 1$ satisfying $k \geq 2B/(\theta d)$, let*

$$\rho = \theta k, \quad H_\theta = \log(1/\theta) - 1 + \theta > 0, \quad A_k(\rho) = \sum_{j=0}^{k} \frac{\rho^j}{j!}, \quad b_j = \left(\frac{\rho^j}{j! A_k(\rho)} \right)^{1/2}.$$

The admissible family

$$f_j(t) = (-1)^j \frac{b_j}{\sqrt{\alpha_j}} Q_j^\rho(t) \chi\left(\sum_i t_i \right) \quad (1 \leq j \leq k), \qquad f_0 = b_0, \qquad f_{k+1} = 0 \tag{5}$$

satisfies

$$\begin{aligned} 1 - q_0 \leq w \leq 1, \qquad & v/\lambda \leq b_k^2 + q_1, \\ \frac{v}{w} \leq \lambda & \frac{(1-\theta)^{-1} e^{-kH_\theta} + e^{-\theta^2 k d^2/(2B^2)}}{1 - e^{-2\theta^2 k d^2/B^2}}, \\ q_0 = e^{-2\theta^2 k d^2/B^2}, \qquad & q_1 = e^{-\theta^2 k d^2/(2B^2)}. \end{aligned} \tag{6}$$

Thus, for a fixed profile and fixed $\lambda, \theta, \beta, \gamma$, v/w decays exponentially in k.

Proof. For fixed k, the support of each positive-dimensional function is contained in $t_i \geq b/\rho > 0$ and $\sum t_i \leq \gamma < \tau$. It is smooth, compact, and symmetric. Also $\sum b_j^2 = 1$, so

$$w = \sum_{j=0}^{k} b_j^2 \mathbb{E}\chi(S_j)^2 \leq 1.$$

Since $j\mu_g/\rho \leq \mu_g/\theta = \beta - d$, Lemma 3.2 gives $\mathbb{P}(S_j > \beta) \leq q_0$ for $1 \leq j \leq k$. The $j = 0$ event is empty. Consequently $w \geq 1 - q_0$.

The coefficient identity

$$b_j = \sqrt{(j+1)/\rho}\, b_{j+1} \qquad (0 \leq j < k) \tag{7}$$

matches exactly the factor from α_j/α_{j+1} and the integration in (4). Therefore, for $0 \leq j < k$,

$$\sqrt{\alpha_j}(f_j + T f_{j+1}) = (-1)^j b_j Q_j^\rho \left[\chi(S) - \frac{1}{\sqrt{\lambda}} \int g(u) \chi(S + u/\rho)\, du \right], \quad S = \sum_i t_i. \tag{8}$$

This includes dimension zero. The expression in brackets vanishes when $S \leq \beta - B/\rho$ and has absolute value at most one everywhere. The first assertion follows pointwise for every $u \in \operatorname{supp} g$. The second follows because each of its two terms lies in $[0, 1]$. The assumption on k gives $B/\rho \leq d/2$; hence the probability of the exceptional set is at most q_1 by Lemma 3.2, uniformly for $1 \leq j \leq k$. It is zero for $j = 0$. Summing the squared norms of (8) costs at most $q_1 \sum_{j<k} b_j^2 \leq q_1$. The remaining top term is at most b_k^2.

For completeness let N have the auxiliary Poisson law of mean $\rho = \theta k$. Then $b_k^2 = \mathbb{P}(N = k)/\mathbb{P}(N \le k)$. Markov gives $\mathbb{P}(N \le k) \ge 1 - \rho/(k+1) \ge 1 - \theta$. The generating function $\mathbb{E}e^{tN} = \exp(\rho(e^t - 1))$ and $t = \log(1/\theta)$ give $\mathbb{P}(N \ge k) \le e^{-kH_\theta}$. These facts prove (6). Finally $H_\theta > 0$ for $0 < \theta < 1$ by the strict inequality $\log x < x - 1$ for $x \ne 1$, applied to $x = \theta$. □

Corollary 4.2 (A fully specified rate)**.** *Take $\theta = 1/2$, $\mu_g < \beta/2$, and $d = \beta - 2\mu_g$. For $k \ge 4B/d$,*

$$\frac{v}{w} \le \lambda \, \frac{2e^{-(\log 2 - 1/2)k} + e^{-kd^2/(8B^2)}}{1 - e^{-kd^2/(2B^2)}}.$$

If $a = \min\{\log 2 - 1/2, d^2/(8B^2)\}$, it is enough to take

$$k \ge \max\left\{ \frac{4B}{d},\ \frac{2B^2 \log 2}{d^2},\ \frac{1}{a}\log\frac{6\lambda}{\varepsilon},\ 1 \right\}$$

to obtain $v/w \le \varepsilon$; a negative last logarithmic threshold imposes no restriction. All thresholds are rounded upward to integers.

Proof. The second threshold makes the denominator at least $1/2$. The numerator is at most $3e^{-ak}$, so $v/w \le 6\lambda e^{-ak}$. □

What this does and does not improve. The source's fixed-profile bound is $v/\lambda \le 2/r + (r/k)^2 + 4s_g^2/(kd^2)$ with $r = \lfloor\sqrt{k}\rfloor$, hence $O(k^{-1/2})$. Our construction removes both flat-window boundaries except the exponentially small top level. It requires a smaller mean profile ($\mu_g < \theta\beta$) and a rescaling $\rho = \theta k$. It does not provide uniform control of B, all derivatives, divisor-sum errors, $\mathbb{E}_X W_X^2$, or the arithmetic threshold X_0. In particular $k = O(\log(1/\varepsilon))$ is a fixed-profile *analytic dimension* statement, not an effective large-prime-gap density formula or a theorem with $k = k(X)$. The auxiliary Poisson law is an exact identity for freely chosen coefficients. It makes no Poisson assertion about primes.

5 A rational taper as an alternative

The project reviewer suggested replacing the source's flat active-level window by a Dirichlet sine taper, and then supplied the rational taper below. We independently prove the resulting estimate. This is a classical discrete-energy device specialized to the source, not a claim of a new discrete spectral principle. Unlike the preceding theorem it keeps the source scale $\rho = k$ and requires only $\mu_g < \beta$.

Proposition 5.1 (Finite rational taper)**.** *Let $1 \le r \le k$, $a = k - r + 1$, and*

$$s_i = i(r + 1 - i)\ (1 \le i \le r), \quad s_0 = s_{r+1} = 0, \quad x_{a+i-1} = \frac{s_i}{(\sum_{\ell=1}^r s_\ell^2)^{1/2}},$$

with $x_j = 0$ off $a, \dots, k$. Choose a fixed profile with $d = \beta - \mu_g > 0$ and use $f_j = (-1)^j x_j Q_j^k \chi(\sum t_i)/\sqrt{\alpha_j}$, including $f_0 = 0$. If $k \ge 2B/d$, then

$$\begin{aligned} w &\ge 1 - e^{-2kd^2/B^2}, \\ \sqrt{v/\lambda} &\le \sqrt{\frac{10}{r^2 + 2r + 2}} + \frac{r}{k} + 2e^{-kd^2/(4B^2)}. \end{aligned} \tag{9}$$

For $r = \lfloor\sqrt{k}\rfloor$ this gives $v/w = O_{\lambda,g,\beta}(k^{-1})$. All squared taper coefficients and its discrete Rayleigh quotient are rational.

Proof. Summation of powers, or direct expansion of $s_{i+1} - s_i = r - 2i$, gives

$$\sum_{i=0}^{r}(s_{i+1}-s_i)^2 = \frac{r(r+1)(r+2)}{3}, \qquad \sum_{i=1}^{r} s_i^2 = \frac{r(r+1)(r+2)(r^2+2r+2)}{30}. \tag{5.1}$$

Extend x by zero. The full uncut normalized residual at level j is $(-1)^j Q_j^k(x_j - \sqrt{(j+1)/k}\, x_{j+1})$, for $0 \le j \le k$. Let $(L_k x)_j = \sqrt{(j+1)/k}\, x_{j+1}$ and $(L_0 x)_j = x_{j+1}$. Their only relevant indices satisfy $j+1 \ge a$. Thus $\|L_k\| \le 1$ and $\|(L_0 - L_k)x\| \le r/k$, since $1 - \sqrt{(j+1)/k} \le 1 - (j+1)/k \le r/k$. The triangle inequality and (5.1) give

$$\|x - L_k x\|_{\ell^2} \le \sqrt{10/(r^2+2r+2)} + r/k.$$

This includes the lower boundary $j = a-1$ and upper boundary $j = k$.

It remains to compare cut and uncut residuals, not just the coefficient sequence. For $\rho = k$, the events $S_j > \beta - B/k$ have probability at most $q = \exp(-kd^2/(2B^2))$, uniformly for $1 \le j \le k$. The change in the direct $x_j Q_j^k$ part has summed squared norm at most $q\sum x_j^2 = q$. In the integrated part, the bracket $\lambda^{-1/2}\int g(u)[\chi(S+u/k)-1]\,du$ vanishes outside the same event and has absolute value at most one. Its summed squared norm is at most $q\sum((j+1)/k)x_{j+1}^2 \le q$. Dimension zero contributes no such error because $B/k \le \beta$. Another triangle inequality gives the additional $2\sqrt{q}$. Finally $w = \sum x_j^2 \mathbb{E}\chi(S_j)^2 \ge 1 - e^{-2kd^2/B^2}$. □

The source's Chebyshev cutoff estimate also suffices here: replace q by $4s_g^2/(kd^2)$ and the mass error by $s_g^2/(kd^2)$, where s_g^2 is the variance under $g^2 du$. The resulting bound still has order k^{-1} at $r \asymp \sqrt{k}$. The compact-support exponential version was also pointed out by the project reviewer; Lemma 3.2 gives a self-contained proof. A sine taper minimizes the unweighted finite Dirichlet energy, but its optimality is not needed for either local conclusion here.

6 Moment coordinates with an error certificate

6.1 The operator and its conditioning

Let

$$\mathcal{H}_k = \mathbb{R} \oplus \bigoplus_{j=1}^{k} L^2_{\mathrm{sym}}(\Delta_j(\tau)), \qquad \|f\|^2_{\mathcal{H}_k} = \sum_{j=0}^{k} \alpha_j \|f_j\|_2^2 .$$

Functions are extended by zero outside the simplex. The target space has the same weighted orthant norms. Define $(Af)_j = Tf_{j+1}$, $(Bf)_j = f_j + Tf_{j+1}$, with the last integrated term zero. The two energies are $w = \|f\|^2_{\mathcal{H}_k}$ and $v = \lambda\|Bf\|^2_{\mathcal{H}_k}$.

Proposition 6.1 (Support-aware approximation budget)**.** *For $f, \widehat{f} \in \mathcal{H}_k$,*

$$\|A\| \le \sqrt{\tau k/\lambda}, \qquad \|B\| \le C_k := 1 + \sqrt{\tau k/\lambda}.$$

If $\left\|\widehat{f} - f\right\|_{\mathcal{H}_k} \le \eta < \sqrt{w(f)}$, then

$$\begin{aligned} w(\widehat{f}) &\ge (\sqrt{w(f)} - \eta)^2, \\ \frac{v(\widehat{f})}{w(\widehat{f})} &\le \lambda\left(\frac{\sqrt{v(f)/\lambda} + C_k\eta}{\sqrt{w(f)} - \eta}\right)^2 . \end{aligned} \tag{10}$$

Proof. For $t \in \Delta_j(\tau)$, Cauchy–Schwarz on the available fiber gives

$$|Te_{j+1}(t)|^2 \le \left(\tau - \sum_i t_i\right) \int_0^{\tau - \sum_i t_i} |e_{j+1}(t,u)|^2 du.$$

Integrating yields $\|Te_{j+1}\|_2^2 \le \tau \|e_{j+1}\|_2^2$, also at $j = 0$. Since $\alpha_j/\alpha_{j+1} = (j+1)/\lambda$, summing proves the operator bound. The reverse triangle inequality for $\widehat{f}$ and the triangle inequality for $B\widehat{f}$ prove (10). □

For the exponentially small target of Theorem 4.1, an approximation of fixed accuracy generally destroys the improvement. To retain a target marked energy of order ε with w bounded below, one needs approximately $\eta = O(\sqrt{\varepsilon}/C_k)$, not just a coordinate reconstruction with small pointwise residual. A Euclidean coefficient error is meaningful only after conversion through the physical Gram matrix.

6.2 Exact finite representations and cross-dimension moments

Let $\Phi_1, \dots, \Phi_N$ be fixed linearly independent admissible smooth families. For $f = \sum_\ell c_\ell \Phi_\ell$ define

$$M_{\ell n} = \langle \Phi_\ell, \Phi_n \rangle_{\mathcal{H}_k}, \qquad H_{\ell n} = \langle B\Phi_\ell, B\Phi_n \rangle_{\mathcal{H}_k}. \tag{11}$$

Then M is positive definite, H is positive semidefinite, and $v/w = \lambda c^T H c/(c^T M c)$ exactly. An invertible change of basis acts by simultaneous congruence and cannot improve this quotient by itself. This is the metric-aware finite moment principle proved in the scoped passages of [5], not a new prime-gap theorem.

The mixed terms in H must be retained. In a single dimension,

$$\langle f_j, Tf_{j+1} \rangle = \int f_j(t) f_{j+1}(t,u)\, dt\, du, \qquad \|Tf_{j+1}\|_2^2 = \int f_{j+1}(t,u) f_{j+1}(t,v)\, dt\, du\, dv.$$

Here the same t occurs in both factors; u and v are separate integration variables. Replacing the second expression by $\int f_{j+1}(t,u)^2 dt\, du$ is false in general. The same-law requirement in [6] applies to these Gram constructions. Positive semidefiniteness of a proposed matrix does not prove that its entries are the correct moments, or that cross-level integrations and boundaries were preserved.

Proposition 6.2 (Finite matrix certificate)**.** *Suppose computed symmetric matrices $\widetilde{M}, \widetilde{H}$ have rigorous operator-norm errors $\left\|\widetilde{M} - M\right\| \le \delta_M$ and $\left\|\widetilde{H} - H\right\| \le \delta_H$, in a stated Euclidean coordinate norm. For any $c \ne 0$ with $D = c^T \widetilde{M} c - \delta_M \|c\|^2 > 0$,*

$$\frac{v(f)}{w(f)} \le \lambda \frac{c^T \widetilde{H} c + \delta_H \|c\|^2}{D}.$$

If these matrices describe $\widehat{f}$ rather than f, the function approximation error must additionally be handled by Proposition 6.1.

Proof. Apply $|c^T(\widetilde{M} - M)c| \le \delta_M \|c\|^2$ and its H analogue. The true numerator is nonnegative and the certified denominator is positive. Division therefore preserves the inequality. □

6.3 An admissible approximation mechanism

The Taylor moment arguments in [7] give a concrete way to pay the approximation error rather than assume it away. For each positive dimension, choose finitely many smooth nonnegative partition functions ψ_ν supported inside the simplex, with $\sum_\nu \psi_\nu \le 1$ everywhere and equal to one on $\operatorname{supp} f_j$. Each support lies within distance h_ν of c_ν; the joining segments lie in a neighborhood where f_j is smooth. Let $P_{\nu,p}$ be its degree-p Taylor polynomial there, and assume the operator norm of $D^{p+1} f_j$ on those segments is at most $D_{\nu,p+1}$. Then

$$\widehat{f}_j = \sum_\nu \psi_\nu P_{\nu,p}, \qquad |f_j - \widehat{f}_j| \le \sum_\nu \psi_\nu \frac{D_{\nu,p+1} h_\nu^{p+1}}{(p+1)!}.$$

Indeed $f_j = (\sum \psi_\nu) f_j$ globally, including points where $f_j = 0$, and the one-dimensional integral Taylor remainder along each segment proves the bound. If the right-hand side is at most e_j on a union of volume V_j, then $\left\| f_j - \widehat{f}_j \right\|_2 \le e_j \sqrt{V_j}$. Symmetrizing $\widehat{f}_j$ over coordinate permutations does not increase this error because the target is symmetric and permutation averaging is an orthogonal projection. The result remains smooth and compactly supported inside the simplex. The scalar f_0 can be kept exact.

The total certified error is $\eta^2 \le \sum_{j=1}^k \alpha_j e_j^2 V_j$. The finite functions $\psi_\nu t^a$ are genuine features, but their weighted and mixed integrals must be evaluated or enclosed. Degree-p products require moments through degree $2p$ together with the relevant cutoff factors and cross-level integrals. Raw monomials on the entire orthant are not admissible source functions. Step-function probability fixtures in the tests below verify algebra, not smooth-source admissibility. Finite-dimensional unisolvence, conditional integration, and source-normalized kernels must not be conflated.

7 A finite weighted-to-prime density ledger

Proposition 7.1 (Finite removal of weights)**.** *On any probability space let $W, V_J, V_I \ge 0$, with $V_I = 0$ or $V_I \ge 1$. Suppose*

$$\mathbb{E}(WV_J) \ge A \ge 0, \quad \mathbb{E}(WV_J^2) \le B_0 > 0, \quad \mathbb{E}(WV_I) \le E_0, \quad \mathbb{E}W^2 \le M_0 > 0.$$

Then, with $q = (A^2/B_0 - E_0)_+$,

$$\mathbb{P}(V_J > 0,\ V_I = 0) \ge q^2/M_0. \tag{12}$$

No independence and no normalization $\mathbb{E}W = 1$ are required.

Proof. Weighted Cauchy–Schwarz gives $(\mathbb{E}WV_J)^2 \le \mathbb{E}(WV_J^2)\mathbb{E}(W\mathbf{1}_{\{V_J>0\}})$. Since $\mathbf{1}_{\{V_I>0\}} \le V_I$, the weighted mass of the good event is at least $A^2/B_0 - E_0$, and is nonnegative. An ordinary Cauchy–Schwarz application bounds this mass by $(M_0 \mathbb{P}(\text{good}))^{1/2}$, proving the result. □

For use with (3), a finite arithmetic error ledger may take $A = \lambda - e_J \ge 0$, $B_0 = C_* + e_2 > 0$, $E_0 = v/w + e_I$, and an independently established bound M_0 on $\mathbb{E}_X W_X^2$. Neither the analytic norm error in (10) nor a small computed value of v supplies these arithmetic errors or M_0.

Proposition 7.2 (Bounded multiplicity and index density)**.** *Let $\lambda > C > 0$. Suppose that for every sufficiently large integer X, at least $\delta_X X$ integers $X < m \le 2X$ have $V_J(m) > 0$ and $V_I(m) = 0$, where $\liminf_{X\to\infty} \delta_X \ge \delta > 0$. Then*

$$\liminf_{N\to\infty} \frac{1}{N} \#\{n \le N : d_n > C \log p_n\} \ge \frac{\delta}{2\lambda}. \tag{13}$$

Proof. For a good m, let p be the largest prime in $(m, m+h]$. There is no prime in $(p, m+2h]$, so the next prime p^+ satisfies $p^+ - p > m + 2h - p \geq h$. A fixed p arises only from $p - h \leq m \leq p - 1$, at most h integer choices, regardless of the full gap length. Consequently there are at least $\delta_X X/h$ distinct such prime gaps, with $X < p \leq 2X + h$. Uniformly in this range $h/\log p \to \lambda > C$.

Fix any $a > 0$ and take $X = \lfloor p_N/(2+a) \rfloor$. For every sufficiently large N, $2X + h < p_N$, so all selected initial primes are indexed at most N. The prime number theorem gives $X/(hN) \to 1/((2+a)\lambda)$. The lower limit is therefore at least $\delta/((2+a)\lambda)$. Letting the fixed a decrease to zero proves (13); no uniform arithmetic estimate in a moving a is used. □

Corollary 7.3 (Conditional assembly with constants)**.** *Assume the source contract* (3) *for a fixed family with $w > 0$, put $D = \lambda^2/C_* - v/w$, and suppose $D > 0$. Writing $M = \limsup_{X\to\infty} \mathbb{E}_X W_X^2 < \infty$, one has*

$$\liminf_{N\to\infty} N^{-1} \#\{n \leq N : d_n > C \log p_n\} \geq \frac{D^2}{2\lambda M} \qquad (0 < C < \lambda).$$

Any strictly smaller positive number is a valid eventual lower-density constant in (1)*.*

Proof. The normalization has $\mathbb{E}_X W_X \to 1$, hence $M \geq 1$. Apply Proposition 7.1 with finite upper and lower bounds converging to the quantities in (3). This gives integer-start density lower limit at least D^2/M. Proposition 7.2 finishes the argument. □

This retains the source's valid counting mechanism and merely avoids unnecessary fixed slack in its constants. Decreasing v/w increases D with other quantities fixed, but a new family can increase M. Thus the improved analytic cancellation alone does not prove an improved numerical value of the actual-prime density.

8 The arithmetic boundary is still substantive

The complete moments file was read, including its proof rather than only Proposition `prop:weight-moments`. The following account identifies the chains used and the obligations not discharged by our finite-dimensional constructions.

Uniform mixed products, source lines 1–278. The general statement fixes a finite list of divisor factors with individual orthant support budgets ρ_r, and a pattern identifying their coordinates with t distinct numerical shifts. Shifts may range over $|a_i| \leq D(\log X)^B$ for fixed D, B. The total support budget is $\sum_r \rho_r < 1$ for an unmarked average, and less than $1/2$ with one distinct prime mark. Dimension-zero factors cost no support. The main term is a singular series times a finite analytic constant, with an additive error of shape $L^{-t-1/2}(\log L)^A$. This is not a relative error divided by the singular series, which can be zero.

Divisor expansion gives CRT-compatible residues and moduli $q \leq X^{\sum \rho_r}$. The budget is a sum over factors, not a sum over all individual dimensions. For an external prime mark, nonreduced residues cannot contain the large prime in question. The source pays endpoint shifts uniformly by a pointwise boundary estimate and removes prime powers before applying a weighted form of Bombieri–Vinogradov. Its divisor multiplicities are absorbed by Cauchy–Schwarz, Euler bounds, and the available logarithmic savings. The distribution theorem and its permissible modulus range are arithmetic inputs, not consequences of our feature coordinates.

Fourier and Euler products, source lines 279–425. Compact smooth extensions provide rapidly decreasing transforms. Absolute convergence first permits divisor expansion and Fourier interchange. Local factors separate generic distinct residues from collisions. After the zeta-type factors are extracted, evaluation at the origin produces the singular series, including its possible zeros. The source compares products additively along a segment and splits small and large primes; it does not divide by a possibly vanishing local factor. Singleton coordinate fibers evaluate a complete derivative on a zero face and vanish. Double fibers yield the Laplace integral responsible for the physical L^2 norm. Fibers of larger multiplicity in the fourth moment use a more general finite analytic constant, not an unjustified pair-only formula.

Averaging the singular series, source lines 426–512. The source truncates at a slowly growing prime bound and uses the primorial residue model only to compute the mean of that *truncated singular series*. It controls the residue discrepancy, removes repeated shifts, and estimates the remaining prime tail using the product of shift differences. This local auxiliary residue calculation is not an identification of the actual primes with independent residue exclusions. Nonnegative truncated products and an additive comparison preserve zero cases. For a fixed pattern the result is an average $1 + o(1)$ over the relevant shift boxes.

Squares and the detector, source lines 513–618. Distinct subsets in Z^2 have unmatched coordinates and vanish in the main term; matching subsets give $\binom{h}{j}/L^j \to \lambda^j/j! = \alpha_j$. A prime outside the building block adds a factor λ. A prime inside it first invokes the exact deletion identity, including the scalar term, and gives the adjacent-dimension form v. The fourth moment is bounded for each fixed finite family.

For detection, $\vartheta(m+a)/L$ is bounded by a fixed smooth one-dimensional divisor-square detector, with the harmless factor $\log(2X + 2h)/L \to 1$. The diagonal prime-count term supplies λw. The two distinct detector positions in the off-diagonal term are treated with an *unmarked* product of six divisor factors, of total support at most $6\tau = 3/4 < 1$. This gives $\lambda^2 c_G^2 w$ and does not assert a two-prime distribution theorem. The ordinary square uses $2\tau < 1$, the marked square $2\tau < 1/2$, and the fourth moment $4\tau < 1$. These inequalities are unchanged by our increase in maximum dimension, because all functions retain the same total simplex support.

Retained details and corrections. No mathematical correction to the source is established by this report. In particular we retain its scalar level, lack of a dimension multiplier in deletion, face condition on complete derivatives, additive zero-safe singular-series error, uniform detector constant, six-factor detector argument, and bounded multiplicity count. The flat-window estimate is valid but locally weaker than the alternatives proved here. A conditional use of (3) is not independent certification of that entire 618-line analytic proof and all its external inputs.

9 Comparison, costs, and further work

9.1 Current public context

A targeted primary-source literature refresh was performed on October 8, 2026 using public titles and public URLs only. Bazzanella–Languasco–Zaccagnini [2] distinguish integer-start intervals from prime-start intervals and give positive-proportion results in specified ranges, with both unconditional and conditional statements. Their abstract does not justify replacing those ranges by every $C > 0$. Jha's current version [3], revised September 12, 2026, describes conditional Poisson behavior

under strengthened Hardy–Littlewood hypotheses in slowly growing ranges and breakdown at larger scales. We use it only as a warning about quantifiers, not as an unconditional input for (1). Granville–Koukoulopoulos–Maynard [4] study how smoothing and moment order affect the arithmetic contribution to divisor-sum moments. This is relevant to the uncontrolled derivative and fourth-moment costs here, but supplies no missing uniform estimate for our growing families.

Those three primary abstract/version records were inspected; their full proofs were not audited for this report. The assigned public manuscript itself was read from the authenticated local build. The search is neither an exhaustive priority review nor a claim that a local construction proves a previously open global theorem. Private titles, passages, paths, and draft formulas were not submitted to public search services.

9.2 Advantages and costs

The main advantage is structural: the exact recurrence (7) eliminates all interior uncut residuals and the lower boundary at once, using an already permitted f_0. Compact support then gives a genuine exponential estimate, with all constants visible. The rational taper offers a less aggressive alternative and exact finite algebra without optimizing a spectrum. The perturbation certificate measures error in the actual cross-dimension norm, and the density ledger states precisely what additional arithmetic estimates would be needed.

The costs are substantial. The small-mean profile can have enormous B/d and large derivatives. Coefficients of the unweighted f_j can be large despite bounded weighted mass. There are many high-dimensional features and mixed moments, and positivity alone does not certify them. Shrinking v/w may greatly increase the fourth-moment constant, so the ordinary-density lower bound need not improve. Neither numerical smallness nor an exact finite coefficient identity proves cancellation of the prime sums at a specified X. No assertion of optimality among all admissible families, or of a new historical cancellation principle, is made.

9.3 Feasible next questions

The immediate analytic optimization problem is to minimize the joint cost of H_θ^{-1}, $B^2/(\theta^2 d^2)$, derivative norms, and the fourth-moment bound, not merely the marked Rayleigh quotient. A second task is rigorous quadrature for the cross-level Gram entries with error small enough for (10). A third is to track the constants in the imported mixed-moment theorem for these particular families. Only after such control could one investigate a slowly increasing $k(X)$ or effective prime-density constants. The present order is: choose C, then $\lambda > C$, then a detector and an admissible finite family, and only then let $X \to \infty$.

10 Evidence and contribution assessment

The companion executable `M052-exact-checks.py` uses exact rational arithmetic for level recurrences, taper energies, two-law finite cutoff fixtures, support-aware perturbation inequalities, prime-deletion subset identities, and finite weighted detection. Its JSON records category counts and the tested script hash. These are independently executable local checks, not samples of an asserted global theorem, a simulation of actual-prime density, or a replacement for the proofs above. Probability fixtures with atoms and piecewise linear cutoffs deliberately test the algebra only; the source-admissible smooth family is established analytically. Parent checks are not added to author-run totals.

The read ledger records complete build reading and the explicitly limited private proof reading, with unread complements. A separate proof ledger marks exact source identities, locally proved improvements, and imported arithmetic. The final PDF is rendered and every page visually inspected; compilation, exact checks, and file hashing do not independently certify all source arithmetic. The final freeze fixes the report and its owned evidence without freezing mutable project-wide review files.

The applicability rating remains L3. The contribution rating is C3: the exponential fixed-profile dimension bound is a notable realized local improvement over the source's flat-window estimate, with a rational polynomial-rate alternative and usable perturbation and density certificates. The classical origins of the elementary devices, the reviewer-supplied taper suggestion, and the imported arithmetic boundary are explicit. C3 is not a historical novelty claim, a certification of (1), or a rating of the entire private corpus.

References

[1] *Positive lower density of large prime gaps*, September 25, 2026. Assigned public build, pinned repository revision `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Public source directory. Source file labels and literal ranges are specified in this report.

[2] D. Bazzanella, A. Languasco, and A. Zaccagnini, *Prime numbers in logarithmic intervals*, Trans. Amer. Math. Soc. 362 (2010), 2667–2684. arXiv:0809.2967. Primary abstract and metadata read; no full-proof audit.

[3] A. Jha, *The Poisson Tail Conjecture for primes in short intervals*, arXiv:2605.23014v2, September 12, 2026. Primary abstract and revision record read.

[4] A. Granville, D. Koukoulopoulos, and J. Maynard, *Sieve weights and their smoothings*, arXiv:1606.06781v4, April 8, 2020. Primary abstract and revision record read.

[5] M. Subbarao, private original `TR-rev-1-old-tr250/TR-rev-1-old-tr250.tex`, lines 121–181 and 455–477. Finite reconstruction, physical metrics, and Gram unisolvence proofs. Local confidential source; not publicly uploaded.

[6] M. Subbarao, private original `TR-core-pat-16/TR-core-pat-16.tex`, lines 216–260 and 305–328. Common-law feature moments, positive semidefiniteness, and remainder control. Local confidential source; not publicly uploaded.

[7] M. Subbarao, private original Ram Calculus, volume 1, revision 2, `ch03-local-moments-rev2.tex`, lines 14–91 and 163–226. Taylor moment proofs and normalization distinctions. Local confidential source; not publicly uploaded.